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| **6.03 Impulse and Momentum** | | | |
| 6.03a<br>6.03c<br><br>6.03b<br>6.03d | Linear momentum | a) Recall and be able to use the definition of linear momentum in one dimension.<br><br>b) Understand and be able to apply the principle of conservation of linear momentum in one dimension applied to two particles.<br><br>*Includes using the formula* $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$. | c) Recall and be able to use the definition of in two dimensions including the vector fo<br><br>d) Understand and be able to apply the prin conservation of linear momentum in two applied to two particles.<br><br>*Includes using the vector form* $m_1\mathbf{u}_1 + m_2\mathbf{u}_2 = m$ |
| 6.03e<br><br>6.03f<br>6.03g<br><br><br><br><br><br><br><br><br><br><br><br>6.03h | Impulse | e) Understand and be able to use the concept of the impulse imparted by a force.<br>f) Be able to use the relationship between the instantaneous impulse of a force and the change in momentum $(I = mv - mu)$.<br><br>*The instantaneous impulse is the impulse associated with an instantaneous change in velocity.*<br><br>*Learners will only be required to apply this to instantaneous events in one dimension.*<br><br>*e.g.*<br>*The direct impact of two smooth spheres.*<br>*An impulsive force acting in the direction of an inelastic string.*<br><br>*Questions involving collision(s) between particles may include multiple collisions and the conditions under which further collisions occur.* | <br><br>g) Understand and be able to apply the impu momentum principle in two dimensions i vector form $\mathbf{I} = m\mathbf{v} - m\mathbf{u}$.<br><br>*e.g.*<br>*The oblique impact of two smooth spheres.*<br>*A smooth sphere with a fixed plane surface.*<br>*An impulsive force acting at an angle to an inela*<br><br><br><br><br>h) Understand and be able to apply the impu momentum principle for a constant force force × time or for a variable force in one only as $\int F\,\mathrm{d}t$. |

| OCR Ref. | Subject Content | Stage 1 learners should… | Stage 2 learners should additionally… |
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| 6.03i<br><br><br>6.03j<br><br><br><br><br><br><br>6.03k<br>6.03l | Restitution | i) Recall and be able to use the definition of the coefficient of restitution, including $0 \le e \le 1$.<br>[*Superelastic collisions are excluded.*]<br>j) Understand and be able to use the terms "perfectly elastic" $(e = 1)$ and "inelastic" $(e = 0)$ for describing collisions.<br>*Learners should know that for perfectly elastic collisions there will be no loss of kinetic energy and for inelastic collisions the bodies coalesce and there is maximum loss of kinetic energy.*<br>k) Recall and be able to use Newton's experimental law in one dimension for problems of direct impact.<br>e.g. *Between two smooth spheres* $(v_1 - v_2 = -e(u_1 - u_2))$ *and a smooth sphere with a fixed plane surface* $(v = -eu)$*, where* $u$ *and* $v$ *are the velocities before and after impact.* | l) Extend their knowledge to problems involving Newton's experimental law in two dimensions.<br>*e.g. The oblique impacts of two smooth spheres and a smooth sphere with a fixed plane surface.*<br>*Questions may involve the velocity expressed as a two dimensional vector.* |